

ASSIGNMENT – 1

Basic Mathematics

B.Sc (Hons.) Agriculture / B.Tech Biotechnology

Number System

Beginner Level

Q1. Define:

- (a) Rational Numbers
- (b) Irrational Numbers
- (c) Real Numbers

Q2. Simplify:

$$\sqrt{50} + \sqrt{18}$$

Q3. Solve:

$$|2x - 5| = 9$$

Intermediate Level

Q4. Prove that:

$$\sqrt{2}$$

is irrational.

Q5. Simplify:

$$(3 + \sqrt{5})(3 - \sqrt{5})$$

Q6. Solve:

$$x^2 - 5x + 6 = 0$$

Advanced Level

Q7. Find all real roots:

$$x^4 - 5x^2 + 4 = 0$$

Q8. Prove:

$$\sqrt{3} + \sqrt{5}$$

is irrational.

Set

Beginner Level

Q1. Define:

- (a) Set
- (b) Subset
- (c) Universal Set

Q2. If:

$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 4, 5, 6\}$$

Find:

- (a) $A \cup B$
- (b) $A \cap B$
- (c) $A - B$

Intermediate Level

Q3. Draw Venn diagrams for:

- (a) $A \cup B$
- (b) $A \cap B$

Q4. Verify De Morgan's Laws.

Advanced Level

Q5. In a class of 100 students:

- 60 like Mathematics
- 45 like Physics
- 25 like both

Find:

- (a) Students liking only Mathematics
- (b) Students liking neither subject

Functions

Beginner Level

Q1. Define function with examples.

Q2. Find domain of:

$$f(x) = \sqrt{x-2}$$

Q3. Evaluate:

$$f(x) = x^2 + 2x + 1$$

Find:

$$f(2)$$

Intermediate Level

Q4. Determine whether:

$$f(x) = 2x + 1$$

is one-one.

Q5. Sketch graph of:

$$y = x^2$$

Advanced Level

Q6. Find inverse of:

$$f(x) = \frac{2x-3}{5}$$

Q7. Discuss injective and surjective properties of:

$$f(x) = x^3$$

Limits and Continuity

Beginner Level

Q1. Define limit.

Q2. Evaluate:

$$\lim_{x \rightarrow 2} (x^2 + 3x)$$

Intermediate Level

Q3. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

Q4. Define continuity.

Advanced Level

Q5. Check continuity of:

$$f(x) = \begin{cases} x + 2, & x < 1 \\ 3x - 1, & x \geq 1 \end{cases}$$

Q6. Explain Left-Hand Limit and Right-Hand Limit with examples.

Extra Questions: Limits and Continuity

Beginner Level

Q1. Define:

- (a) Limit of a function
- (b) Continuity
- (c) Discontinuity

Q2. Evaluate:

$$\lim_{x \rightarrow 3} (x^2 + 2x)$$

Q3. Evaluate:

$$\lim_{x \rightarrow 1} (2x + 5)$$

Q4. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

Q5. Evaluate:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$$

Q6. Check continuity of:

$$f(x) = x^2 + 1$$

Q7. Explain Left-Hand Limit (LHL).

Q8. Explain Right-Hand Limit (RHL).

Intermediate Level

Q9. Evaluate:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

Q10. Evaluate:

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$$

Q11. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\tan x}{x}$$

Q12. Check continuity of:

$$f(x) = \begin{cases} x + 1, & x < 2 \\ 3x - 1, & x \geq 2 \end{cases}$$

Q13. Define removable discontinuity with example.

Q14. Define jump discontinuity with example.

Q15. Define infinite discontinuity with example.

Q16. Define oscillatory discontinuity with example.

Q17. Determine the type of discontinuity:

$$f(x) = \frac{x^2 - 1}{x - 1}$$

Q18. Determine continuity at:

$$x = 0$$

for:

$$f(x) = \frac{\sin x}{x}$$

Extra Questions: Limits and Continuity

Advanced Level

Q1. Evaluate:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

Q2. Evaluate:

$$\lim_{x \rightarrow \infty} \frac{2x^2 + 3}{x^2 - 1}$$

Q3. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$$

Q4. Check continuity of:

$$f(x) = \begin{cases} x^2 + 1, & x < 1 \\ 2, & x = 1 \\ 2x - 1, & x > 1 \end{cases}$$

Q5. Discuss continuity of:

$$f(x) = |x|$$

Q6. Find values of:

$$k$$

for continuity:

$$f(x) = \begin{cases} kx + 1, & x < 2 \\ 5, & x = 2 \\ x + k, & x > 2 \end{cases}$$

Q7. Determine type of discontinuity:

$$f(x) = \tan x$$

Q8. Determine points of discontinuity:

$$f(x) = \frac{1}{x^2 - 4}$$

Q9. Prove:

$$\lim_{x \rightarrow 0} \frac{\sin ax}{x} = a$$

Q10. Solve:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$$

Submission Instructions:

1. Write answers neatly.
2. Show all mathematical steps.
3. Mention Name, Roll Number and Course.
4. Submit handwritten PDF.
5. Attempt all Beginner questions compulsory.